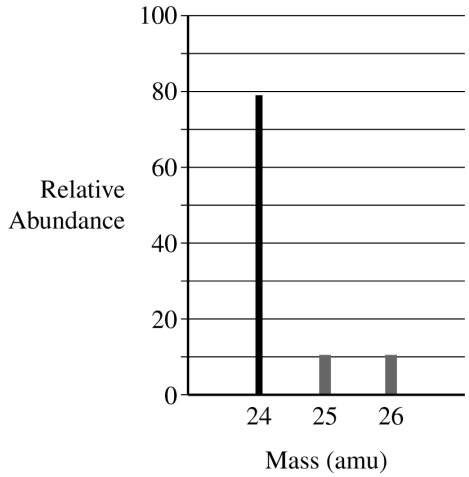
Question 1: Long Answer — 10 points

A (i) For the correct plotted lines: **Point 01**

The abundance for the two lines should be between 10 and 11.



(ii) For the correct answer: **Point 02**

Magnesium-26 has one more neutron than magnesium-25 does.

B (i) For a correct explanation: **Point 03**

The charge on the sodium ion is less than the charge on the magnesium ion. A smaller charge results in a weaker Coulombic attraction between Na^+ *and water.*

(ii) For a correct explanation: **Point 04**

The Na^+ *ion is larger than the* Mg^{2+} *ion, so the distance between the* Na^+ *and the oxygen on the water molecule will be greater. As distance increases, Coulombic attraction decreases.*

C For the correct calculated value: **Point 05**

$$\text{pOH} = -\log(2.80 \times 10^{-4}) = 3.553$$

$$\text{pH} = 14 - \text{pOH} = 14 - 3.553 = 10.447$$

D For the correct calculated value: **Point 06**

$$M_1V_1 = M_2V_2$$

$$M_2 = \frac{(1.85 \times 10^{-3}\,M)(0.03500\text{ L})}{(0.03500\text{ L} + 0.05000\text{ L})} = 7.62 \times 10^{-4}\ M$$

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E	(i) For the correct expression: $K_{sp} = [Mg^{2+}][OH^-]^2$	**Point 07**
	(ii) For the correct calculated value, consistent with part D and part E (i): $Q = [Mg^{2+}][OH^-]^2 = (7.62 \times 10^{-4})(1.65 \times 10^{-4})^2 = 2.07 \times 10^{-11}$	**Point 08**
	(iii) For the correct prediction and justification, consistent with part E (ii). Examples of acceptable responses may include the following: • $Q > K_{sp}$, *so a precipitate will form.* • *The concentration of the ions in solution (represented by Q) is greater than that of a saturated solution, so a precipitate will form.*	**Point 09**
F	For the correct answer and justification: *Decrease. The* H^+ *from* $HNO_3(aq)$ *will react with* OH^-, *decreasing* $[OH^-]$ *and causing* $Q < K_{sp}$. *As a result, more* $Mg(OH)_2(s)$ *will dissolve until equilibrium is reestablished, resulting in less* $Mg(OH)_2(s)$.	**Point 10**

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